

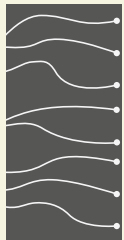
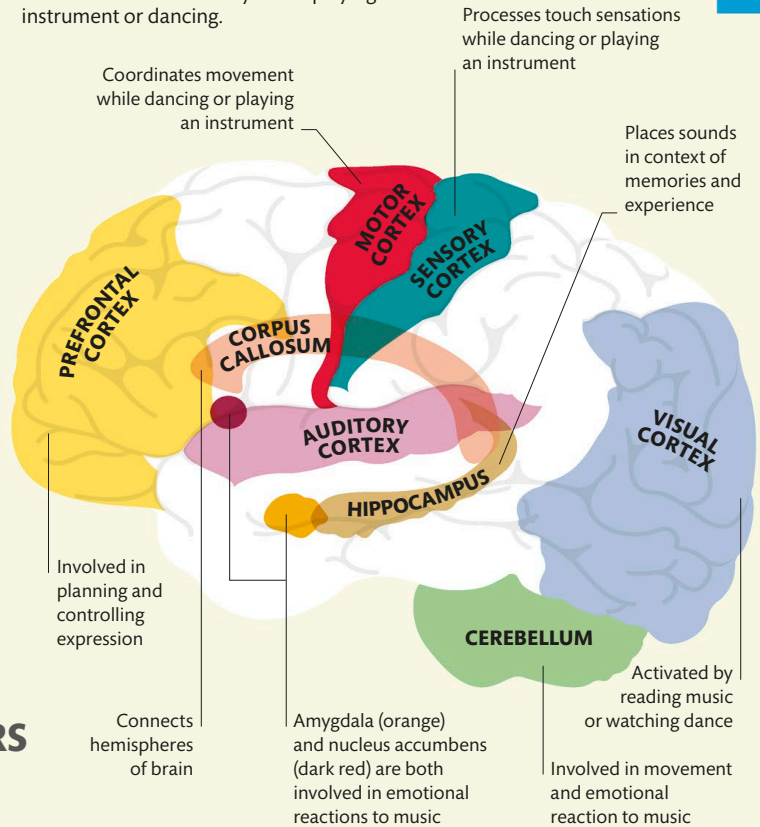


Music and the brain

Music engages many areas of the brain. In addition to processing the sounds, listening to music triggers the memory and emotion centers in the brain, while recalling lyrics involves the language centers. Performing music makes even greater demands: the visual cortex is stimulated by reading music, the frontal lobe is involved in planning actions, and the motor cortex coordinates movement. Musicians are known to have a greater ability to use both hands because music requires coordination of motor control, somatosensory touch, and auditory information. Unlike listeners, who process music in the right hemisphere, professional musicians use the left. They also have a thicker corpus callosum (the region linking the two hemispheres) and tend to have larger auditory and motor cortices.

Mapping music

Scans show that several areas of the brain are active when listening to music, and even more are involved when you are playing an instrument or dancing.



30,000
THE NUMBER OF FIBERS
THAT MAKE UP THE
AUDITORY NERVE

HIGHS AND LOWS

Humans can detect a good range of frequencies, but other animals can hear things far beyond our limits. Animals such as bats and dolphins use high frequencies in echolocation, while elephants and whales produce low rumbles that can travel long distances. Humans are most sensitive to frequencies between 2 kHz and 5 kHz, which do not require great intensity to be heard. Young people have the best hearing range, from 20 Hz to 20 kHz, but there is a gradual loss of higher frequencies with age, with older people having a limit of around 15 kHz.

